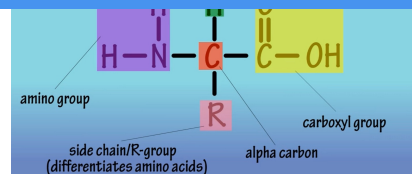
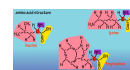
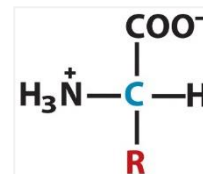


Amino acids

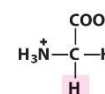


- **Amino acids** are the simplest units of a protein molecule and they form the building blocks of protein structure.
- Each amino acid contains **two** functional groups, an **amino (-NH₂)** group and **carboxyl (-COOH)** group.
- All of the amino acids (except **proline**) have **free** carboxyl group and an amino group.
- In all amino acids found in proteins, those groups (**amino group**, **carboxyl group** and a **hydrogen atom**) are attached to the carbon atom, this type is called **alpha (α) amino acids**.
- The unique characteristics of the (20) amino acids are due to a side chain (R) group, which can be an **alkyl, hydroxyl, aromatic, or heterocyclic group**.
- At neutral pH the amino acids exist as **zwitterions (hybrid)**, due to the **negatively charged carboxyl group** and a **positively charged amino group**.
- All the amino acids have trivial or common names based on the source from which they were first isolated or based on their properties. For example: Asparagine was named so, as it was isolated from **asparagus** and glycine named because of its sweet taste.

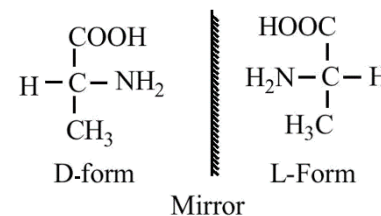


Optical properties of amino acids

- Because the **α-carbons** of 19 of the 20 amino acids are attached to four different groups (i.e., a hydrogen, a carboxyl group, an amino group, and an R group), they are referred to as **asymmetric**, or chiral carbons.
- **Glycine** is a **symmetrical** molecule because its -carbon is attached to two hydrogens.
- Based on the position of amino group on the asymmetric carbon atom, amino acids exist in two types, they are **Dextro** and **Levo** amino acids.
- The two forms in each pair are termed **stereoisomers** (or **enantiomers**).
- The amino acid having the **NH₂ group** on the **right** is called **D-amino acid**.
- The amino acid having the **NH₂ group** on the **left** is called **L-amino acid**.
- All proteins amino acid are **α amino acids** because the NH₂ group is attached to the α carbon atom which is next to the COOH group.



Glycine



- Amino acid residues in protein molecules are exclusively **L-stereoisomers**.
- Although **D-amino acids** occur in nature, most often in bacterial cell walls and in some antibiotics, they are not found in proteins.
- As the amino acids in proteins are **asymmetric**, the proteins that contain them also exhibit asymmetric properties.

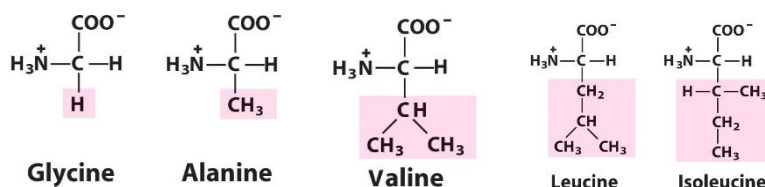
Photogenic amino acids

D, L

Classification of amino acids according to their (R) group;

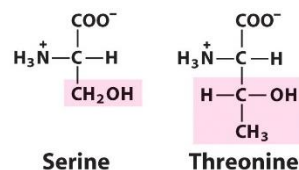
1- Aliphatic amino acids

- **Glycine** (non-polar, *aliphatic R-group*)
- **Alanine** (non-polar, *aliphatic R-group*)
- **Valine** (non-polar, *aliphatic R-group*)
- **Leucine** (non-polar, *aliphatic R-group*)
- **Isoleucine** (non-polar, *aliphatic R-group*)



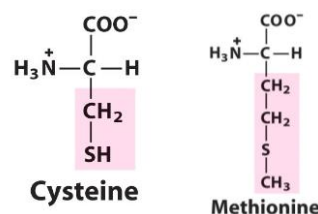
2- Amino acids containing hydroxyl (-OH) group

- **Serine** (polar - uncharged R groups)
- **Threonine** (polar - uncharged R groups)



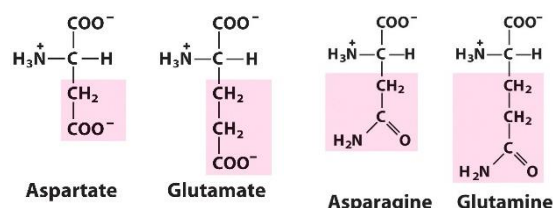
3- Amino acids containing thiol or (-SH) group

- **Cysteine** (polar - uncharged R groups)
- **Methionine** (non-polar, *aliphatic R-group*)



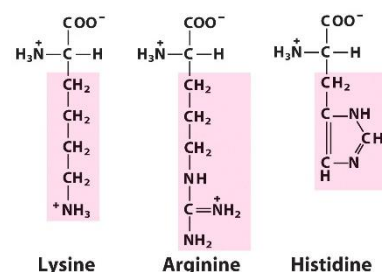
4- Acidic amino acids and its amides

- **Aspartate** (polar - negatively charged [Acidic] R groups)
- **Glutamate** (polar - negatively charged [Acidic] R groups)
- **Asparagine** (polar - uncharged R groups)
- **Glutamine** (polar - uncharged R groups)



5- Basic amino acids

- **Lysine** (polar - positively charged [basic] R-groups)
- **Arginine** (polar - positively charged [basic] R-groups)
- **Histidine** (polar - positively charged [basic] R-groups)



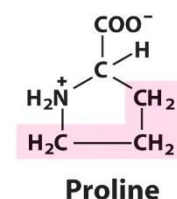
6- Heterocyclic amino acids (Containing imino '-NH-' group)

- **Proline** (non-polar, *aliphatic R-group*)

▪ Differs from other amino acids in that its side chain and α-amino N form a rigid, five-membered ring structure.

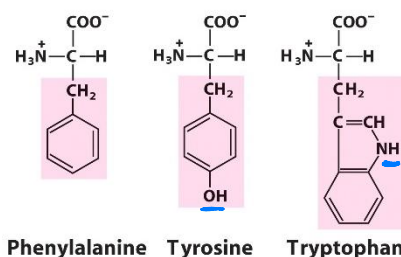
▪ Proline, then, has a secondary (rather than a primary) amino group. Thus proline is more accurately classified as an "α-imino acid."

▪ All of the amino acids except proline have free carboxyl group and an amino group.



7- Aromatic amino acids

- **Phenylalanine** (non-polar, *aromatic R group*)
- **Tyrosine** (non-polar, *aromatic R group*)
- **Tryptophan** (non-polar, *aromatic R group*)



Classification of amino acids according to their polarity;

Non-polar, aliphatic R-group

- The R-group in this class of amino acids are non-polar (or) **hydrophobic** ('water-fearing'), which are **glycine, alanine, valine, proline, leucine, isoleucine** and **methionine**.

Non-polar, aromatic R groups

Polarity

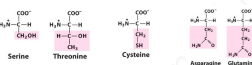
Tyrosine > Tryptophan > Phenylalanine

- Includes **phenylalanine, tyrosine** and **tryptophan**.
- The hydroxyl group of **tyrosine** can form hydrogen bond with other compounds and it is an important functional group in some enzymes. Phenyl alanine is significantly more non-polar than Tyrosine and tryptophan because of the hydroxyl group of tyrosine and the nitrogen in the indole ring in tryptophan.

Polar - uncharged R groups

- The R groups of these amino acids are more soluble in water or more **hydrophilic** ('water attracting'), than those of the non-polar amino acids because they contain functional groups that form hydrogen bonds with water. Includes **serine, threonine, cysteine, asparagine** and **glutamine**.

- The polarity of serine and threonine is contributed by their hydroxyl groups which enables them to participate in hydrogen bonding; that of cysteine by its sulphhydryl (-SH) group; and that of asparagine and glutamine by their amide groups.

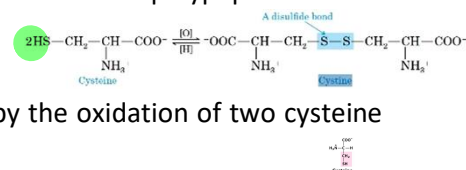


- Because the amide functional group is highly polar, the hydrogen-bonding capability of asparagine and glutamine has a significant effect on protein stability.

- Cysteine** is readily **oxidized** to form a covalently linked **dimeric amino acid called cystine**, in which two cysteine molecules are joined by a **disulfide bond**.

- The disulfide linkage is strongly hydrophobic (nonpolar), plays role in the structures of many proteins by forming covalent links between parts of a protein molecule or between two different polypeptide chains.

Cystine Is a Derived Amino Acid



- A derived amino acid found in many proteins is cystine. It is formed by the oxidation of two cysteine thiol side chains, joined to form a disulfide covalent bond.

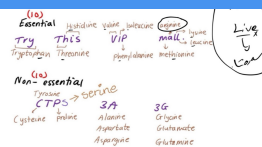
- Within proteins disulfide links of cystine formed from cysteines which is separated from each other in the primary structure, have an important role in stabilizing the folded conformation of proteins.

Polar - Positively charged (basic) R-groups

- The amino acids in which the R groups have significant positive charges at pH 7.0 are **lysine, arginine** and **histidine**.

Polar - Negatively charged (Acidic) R groups

- The two amino acids having R groups with a net negative charge at pH 7.0 are **aspartate** and **glutamate**.



- Based on their inclusion in the diet, amino acids are classified into:

A- Essential amino acids: These are the ones which are not synthesised by the body and must be taken in diet.

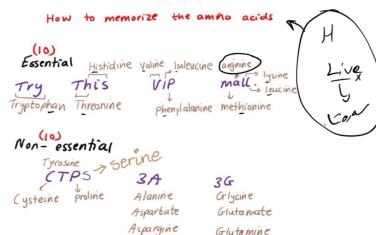
B- Non-essential amino acids: They can be synthesised by the body and may not be the requisite components of the diet.

- The body can synthesize some of these amino acids, but not in sufficient quantities for its needs, especially in the case of growing children (particularly children's requirement for arginine and histidine). So **Histidine** and **arginine** are essential for infants and small children.

C- Semi-essential amino acids: These are growth promoting factors since they are not synthesised in sufficient quantity during growth. They include **arginine** and **histidine**. They become essential in **growing children, pregnancy and lactating women**.

Essential amino acids (Phatih LVL Methionine Threonine)	Non-essential amino acids (Tyrosine GAP GAS GAC)
Phenylalanine, Arginine, tryptophan, Isoleucine, Histidine, Leucine, Valine, Lysine, Methionine, Threonine	Tyrosine, Glycine, Alanine, Proline, Glutamine, Asparagine, Serine, Glutamic acid, Aspartic acid, Cysteine

Live + H



New Amino Acids;

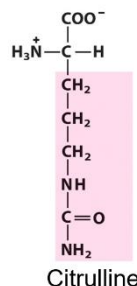
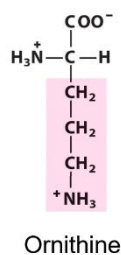
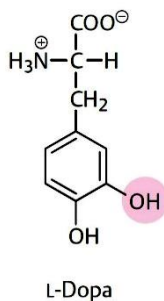
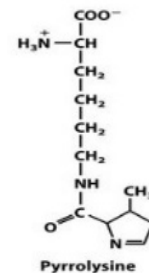
- In addition to 20 L-amino acids that take part in protein synthesis, recently two more new amino acids described. They are:

A. Selenocysteine: 21st amino acids it occurs at the "active site" of several enzymes such as (**Glutathione peroxidase**) and (**Glycine reductase**).

B. Pyrrolysine: 22nd amino acid. The **STOP codon UAG** can code for pyrrolysine.

- Certain amino acids which do not exist in proteins are called **non protein amino acids**.

- For example: - **β-alanine, Ornithine, Citrulline, L-Dopa**.



Uncommon Amino Acids in Protein:

• In addition to the 20 common amino acids, proteins may contain residues created by modification of common residues already incorporated into a polypeptide.

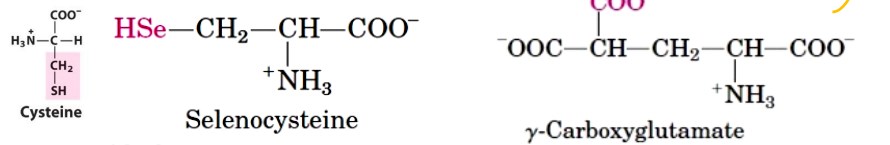
• Among these uncommon amino acids are **4-hydroxyproline**, a derivative of **proline**, and **5-hydroxylysine**, derived from **lysine**. The former is found in plant cell wall proteins, and **both are found in collagen**.

• In many proteins some of the amino acids are modified after their incorporation; e.g., in collagen a hydroxyl group is added to each of several proline residues to yield hydroxyproline, and this does not constitute direct genetic encoding.

• **6-N-Methyllysine** is a constituent of **myosin**, a contractile protein of muscle.

• **γ-carboxyglutamate**, found in the **blood clotting** protein prothrombin.

• **Selenocysteine** is a special case. This rare amino acid residue is introduced during protein synthesis rather than created through a **postsynthetic modification**. It contains **selenium** which **replaced by the sulfur of cysteine**, which is constituent of just a few known proteins.



Biologically Active Amino Acids

• Several amino acids or their derivatives **act as chemical messengers**.

• **For example** \\

• **Gama amino butyric acid** (GABA, a derivative of **glutamate**, an **inhibitory neurotransmitter**, its underproduction is associated with epilepsy).

• **Serotonin** (happiness, neurotransmitter) and **melatonin** is a hormone that is produced by the pineal gland in animals and regulates sleep and wakefulness) are derivatives of **tryptophan**.

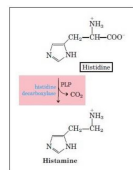
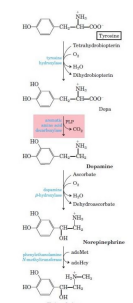
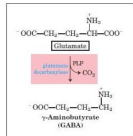
• **Thyroxine** (a **tyrosine** derivative) and **indole acetate** (plant growth hormone) (a **tryptophan** derivative) are hormones chemical signal molecules produced in one cell that regulate the function of other cells.

• **Tyrosine** is a precursor to family of catecholamines that includes **dopamine**, **norepinephrine**, and **epinephrine**. Levels of catecholamines are **correlated with changes in blood pressure**.

• The neurological disorder **Parkinson's disease** is associated with an **underproduction of dopamine**, and it has traditionally been treated by **administering L-dopa**.

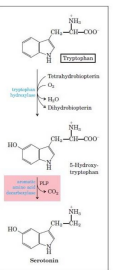
• Amino acids are precursors of a variety of complex nitrogen-containing molecules. Examples include nucleotides and the nucleic acids, **heme**, and **chlorophyll**.

• **Histidine** undergoes decarboxylation to **histamine**, a powerful vasodilator in animal tissues. Histamine is released in **large amounts as part of the allergic response**, and it also stimulates acid secretion in the stomach.



This modification introduces an affinity for calcium ions. In the blood clotting cascade, vitamin K is required for γ-carboxylation of clotting factors II, VII, IX, X and protein Z. (A from biochemistry, in Spanish - "comparaciones")

پریشنی کاری

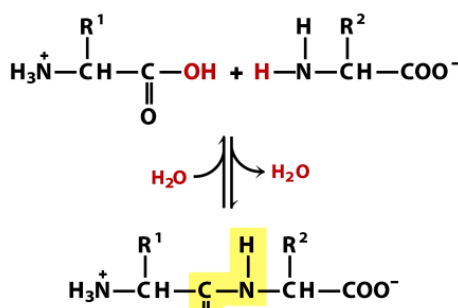


Protein

- About three quarters (**3/4**) of the body solids are proteins.
- All proteins in human are polymers of (20) different amino acids, and they arranged in a specific order that determines the characteristics of the protein and its physiological action.
- In addition to C, H, and O which are present in carbohydrates and lipids, proteins also contain Nitrogen. The **nitrogen** content is around 16% of the molecular weight of proteins. Small amounts of S and P are also present.
- Protein molecules are very large molecules with a high molecular weight. Protein can be broken down into smaller units **by hydrolysis**.
- Proteins are the **unbranched** polymers of L- α -amino acids.

Protein Biosynthesis

- The amino acids can aggregate into long chains by peptide linkages.
- The **peptide bond** is formed between the carboxyl group of one amino acid and the amino group of adjacent amino acid.
- Two amino acid molecules can be covalently joined through a substituted amide linkage, termed a peptide bond, to yield a dipeptide.
- Such a linkage is formed by removal of the molecule of water (dehydration) from the **α**- carboxyl group of one amino acid and the **α**- amino group of another.
- Three amino acids can be joined by two peptide bonds to form a tripeptide; similarly, amino acids can be linked to form tetrapeptides and pentapeptides.
- When a few amino acids are joined in this fashion, the structure is called an oligopeptide.
- When many amino acids are joined, the product is called a polypeptide.
- Proteins may have thousands of amino acid residues.
- Although the terms “protein” and “polypeptide” are sometimes used interchangeably, for molecules composed of over 50 amino acids and the term **peptide is used for molecules of less than 50 amino acids**, polypeptides generally have molecular weights below 10,000, and those called proteins have higher molecular weights.
- Does all polypeptides are protein? **A\ No**, but all proteins are polypeptides.



Although the terms “protein” and “polypeptide” are sometimes used interchangeably, molecules referred to as polypeptides generally have molecular weights below 5000, and those called proteins have higher molecular weights.

Examples on Peptides:

1- Dipeptide (tow amino acids joined by one peptide bond):

Example: **Aspartame** which acts as sweetening agent being used in replacement of cane sugar. It is composed of aspartic acid and phenyl alanine.

2- Tripeptides (3 amino acids linked by two peptide bonds).

Example: **Glutathione (GSH)** which is formed from 3 amino acids: glutamic acid, cysteine and glycine. It helps in absorption of amino acids, protects against hemolysis of RBC by breaking H_2O_2 which causes cell damage.

3- Octapeptides: (8 amino acids)

Examples: Two hormones; oxytocine and vasopressin (ADH).

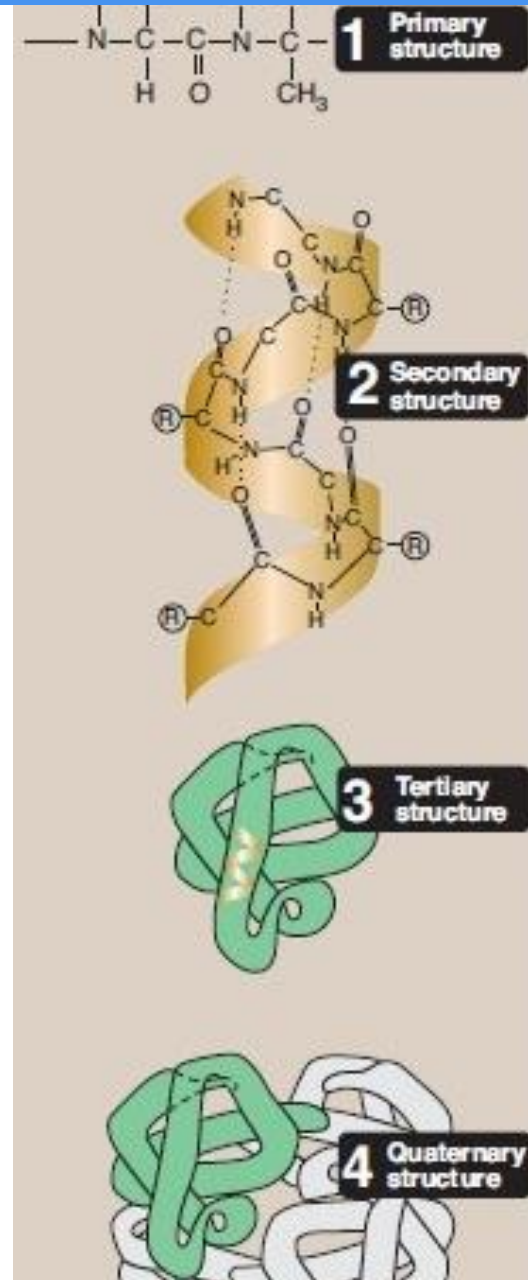
4- Polypeptides: 10- 50 amino acids: e.g. Insulin hormone

Structure of proteins

Linder strom - Lang suggest four types of structural organization for proteins.

They are

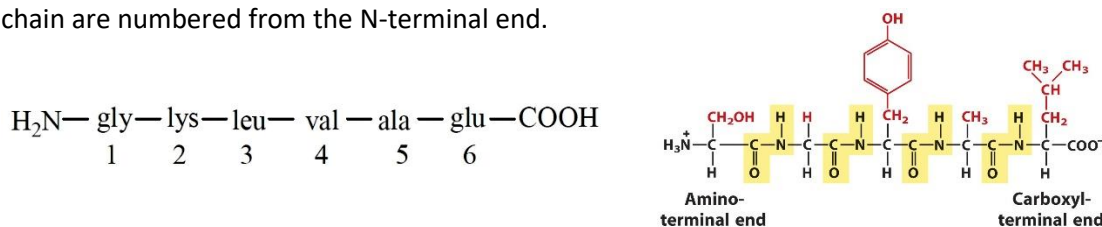
1. Primary structure.
2. Secondary structure.
3. Tertiary Structure.
4. Quaternary structure.



Structure of proteins

1- Primary structure:

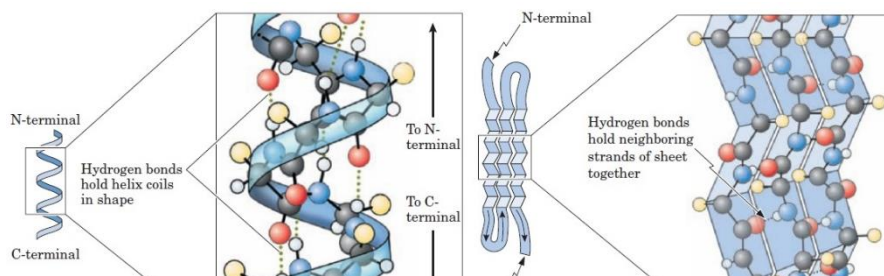
- The primary structure of protein is defined as the sequence of amino acid residues making up its **polypeptide chain**.
- The primary structures of proteins are translation of information contained in genes.
- Each peptide or protein in the body has its own unique (primary structure) sequence of amino acids.
- The protein may be formed from one or more polypeptide chains.
- The amino acid with free NH_3^+ group is known as amino terminal or (N-terminal) amino acid, and the amino acid with free COO^- group is known as the carboxyl or (C-terminal). The amino acids in a polypeptide chain are numbered from the N-terminal end.



- We can say that the primary structure of a protein determines to a large extent the native of secondary and tertiary structures.
- That is, the particular sequence of amino acids on the chain enables the whole chain to fold in such a way as to assume its final shape, without its particular three dimensional shape, a protein cannot function.
- One of the largest natural polypeptide chains in humans is that of **apolipoprotein B100**, which contains **4536 amino acid** residues in one polypeptide chain.
- Chain length alone, however, does not determine the function of a polypeptide.
- Many small peptides with less than ten amino acids perform important biochemical and physiological functions in humans.

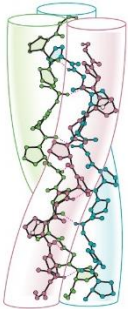
2- Secondary structure:

- The folding part of this chains into regular structure such as **α -helixes** and **β -pleated sheets**, it results from the relationship between amino acids located relatively near to each other in the peptide chain.
- Secondary structure is the result of hydrogen bonding between the amide hydrogen's and carbonyl oxygen's of the peptide bond. Many hydrogen bonds are needed to maintain the secondary structure, the two most common types of secondary structure are the **α -helixes** and **β -pleated sheets** because they maximize the hydrogen bonding in the backbone.



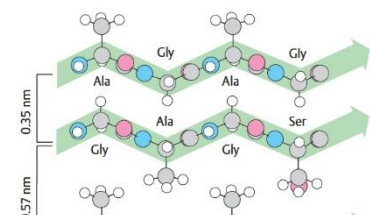
Collagen; [A Protein That Holds Us Together]

- Repeating pattern classified as a **secondary structure** is the extended helix (Triple Helix) of collagen.
- Collagen is rich in **proline** and **hydroxyproline** it forms a **triple helix**.
- The **triple helix** is stabilized by both noncovalent as well as covalent bonds. Interchain hydrogen bonds between different peptide chains are formed which are almost perpendicular to the long axis of α -helix.
- **Collagen** is composed of **three left-handed polypeptide helices** that are twisted around one another to form a **triple helix** (superheli).
- In addition, interchain additional cross links, secondary amide bonds of peptide bonds also are responsible for stabilize the triple helix.
- Collagen is the structural protein of **connective tissues** (bone, cartilage, tendon, blood vessels, skin).
- Collagen, the most abundant protein in humans, it makes up about **30%** by weight of all the body's protein.
- Each strand of collagen consists of repetitive units that can be symbolized as Gly—X—Y; that is, every third amino acid in the chain is glycine.
- Approximately one-third of the amino acids are glycine, Two unusual, hydroxylated amino acids account for nearly one-fourth of the amino acids in collagen. These amino acids are 4-hydroxyproline and 5-hydroxylysine.
- When collagen is synthesized, the amino acids **proline** and **lysine** are incorporated into the chain of amino acids. These are later modified by two enzymes to form **4-hydroxyproline** and **5-hydroxylysine**.
- Both of these enzymes require **vitamin C** to carry out these reactions. In fact, this is the major known physiological function of vitamin C.
- Without hydroxylation, hydrogen bonds cannot form and the triple helix is weak, resulting in fragile blood vessels.



Keratin;

- **Keratin**, a fibrous protein of hair, skin, fingernails, horns, and wool, is a protein that does have a predominantly α -helix structure.
- In contrast to keratin, myoglobin, whose structure is also highly α -helical, is a globular, flexible molecule.
- **α -keratin** is rich in the hydrophobic residues Ala, Val, Leu, Ile, Met, and Phe.
- Silk is made of fibroin (Fibroin is an insoluble protein present in silk created by spiders) The fibroin protein consists of layers of antiparallel beta sheets. Its primary structure mainly consists of the recurrent amino acid sequence (Gly-Ser-Gly-Ala-Gly-Ala)_n.
- The high glycine (and, to a lesser extent, alanine) content allows for tight packing of the sheets, which contributes to silk's rigid structure and tensile strength that exists mainly in the β -pleated sheet form.
- The primary structure of silk contains sections that consist of only (25%) alanine and (42%) glycine.
- Alanine section is responsible for the formation of β -pleated sheets.



3- Tertiary (Ternary) structure:

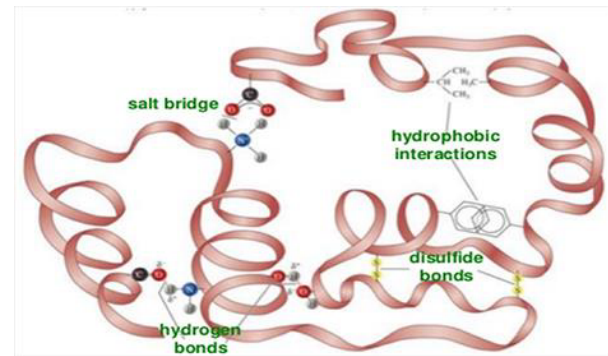
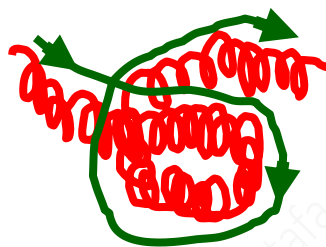
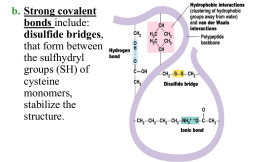
• The polypeptide chain with secondary structure may be further **folded (super-folded)** or **twisted** about itself forming many sizes. Such a structural confirmation is called **tertiary structure**.

• The forces that stabilize the tertiary structure:

1. **Van der waals forces** between **non-polar R group** amino acids that are hydrophobic.
2. **Hydrogen bonds** between the **polar R groups** of the polar amino acid.
3. **Ionic bond** (salt bridge) between the R groups of oppositely charged amino acid.
4. **Covalent** (disulfide) bonds between thiol-containing amino acid.

a. The **weak interactions** include:

- **Hydrogen bonds** among polar side chains
- **Ionic bonds** between charged R groups (basic and acidic amino acids)
- **Hydrophobic interactions** among hydrophobic (non polar) R groups.



4- Quaternary structure:

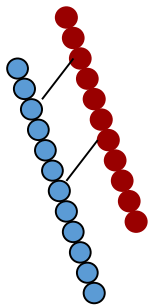
• Some proteins are made up of more than one polypeptide chain. These peptide chains held together by non-covalent interactions or by covalent links, is called **quaternary structure**.

• Each constituent peptide chain is called as a **subunit**. The subunits can be **identical** or **quite different** in their primary, secondary or tertiary structure.

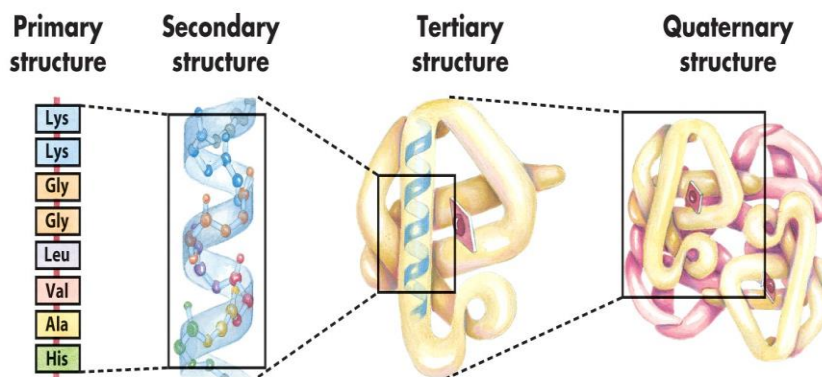
• The forces that hold the quaternary structure are the same as those in the tertiary structure. (**hydrogen bond**, **ionic bridge**, **van der waals forces** and **Covalent or disulfide bridge**).

• In some cases the quaternary structure of a functional protein involves binding to a non-protein group. This additional group is called prosthetic group.

• For example many of the **receptor proteins** on cell surfaces are **glycoproteins**; these are proteins with sugar groups covalently attached.



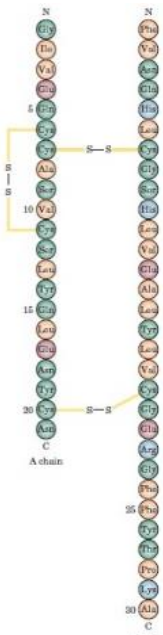
Levels of protein structure



Can a protein or peptide perform the same function with some difference in its sequence?

- An example is the **hormone insulin**. Human insulin consists of two chains having a total of **51 amino acids**. The **two chains are connected by disulfide bonds**. Insulin is necessary for proper utilization of carbohydrates, and people with severe diabetes must take insulin injections. So bovine insulin or insulin from hogs or sheep is used instead of human insulin.

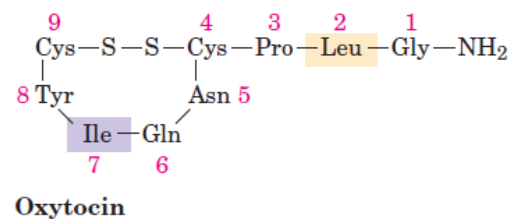
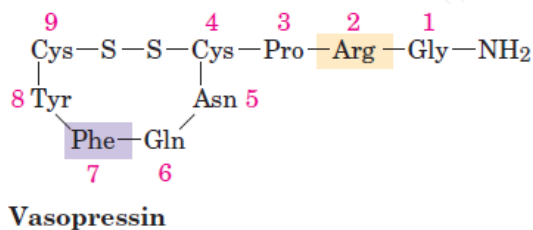
- Insulin from these sources is similar, but not identical to human insulin. The differences are entirely in the 8, 9, and 10 positions of the A chain and the C-terminal position (30) of the B chain. Despite the slight differences in structure, all of these insulins perform the same function and even can be used by humans. However, none of the other three is quite as effective in humans as human insulin. This is one of the reasons that recombinant **DNA techniques are now used to produce human insulin from bacteria**.



	A Chain			B Chain
	8	9	10	30
Human	—Thr—	—Ser—	—Ile—	—Thr
Bovine	—Ala—	—Ser—	—Val—	—Ala
Hog	—Thr—	—Ser—	—Ile—	—Ala
Sheep	—Ala—	—Gly—	—Val—	—Ala

(Amino Acid Sequence Differences for Human, Bovine, Hog, and Sheep Insulin)

- In contrast to the previous examples, some small changes in amino acid sequence make a great deal of difference. Consider two peptide hormones, oxytocin and vasopressin.
- Each of these peptides contains nine amino acid residues
- The difference between these two peptides is that **oxytocin** has an **leucine** residue at **position 2** and an **isoleucine** residue at **position 7**, and **vasopressin** has a **arginine** residue at **position 2** and an **phenylalanine** residue at **position 7**.



- Both of these peptides have considerable physiological importance as **hormones**.
- Oxytocin** induces labor in pregnant women and controls contraction of uterine muscle. Also plays a role in stimulating the flow of milk in a nursing mother.
- Vasopressin** plays a role in the control of blood pressure by regulating contraction of smooth muscle and stimulates reabsorption of water by the kidney, thus having an antidiuretic effect. More water is retained, and the blood pressure increases.
- Another instance where a minor (small) change makes a major difference is in the **blood protein hemoglobin** which is **sickle cell anemia**.

Sickle cell anemia

- **Sickle cell anemia** was the first human disease that was shown to be caused by mutation of a specific protein.
- Patients with sickle cell anemia have a point mutation in their **hemoglobin molecules**.
- Together, the four polypeptides of a hemoglobin molecule contain about 600 amino acids.
- The **β -chains** of normal hemoglobin have a **glutamic acid** residue at **position 6**.
- In the hemoglobin from sickle cell anemia patients, this **glutamic acid** (negatively charged amino acid) is **replaced by a valine** (hydrophobic amino acid), a very nonpolar amino acid.
- Thus with only two amino acid changes out of 600 (one residue per β -chain), the entire biological activity of the protein is severely altered.
- Sickle cell anemia receives its name from the sickled appearance of the red blood cells that form in this conditions.
- The sickled cells are unable to pass through small capillaries of the circulatory system, this results in **damage** of many organs. This malformation occurs in the capillaries.
- As a result of this change in shape, the cells may clog (block out) the capillaries. The body's defenses destroy the clogging cells, and loss of the blood cells causes anemia.
- The U. S. Food and Drug Administration approved **hydroxyurea** as **treatment**.
- **Hydroxyurea** prompts the bone marrow to manufacture **fetal hemoglobin (HbF)**, which **does not have beta chains** where the mutation occurs. Thus red blood cells containing HbF do not sickle and do not clog the capillaries.
- With **hydroxyurea therapy**, the bone marrow still manufactures mutated sickled HbS, but the presence of cells with fetal hemoglobin dilutes the concentration of the sickling cells, thereby relieving the symptoms of the disease.
- **Hemoglobin** is formed of **4 polypeptide chains**, of which two α -chains are of one kind and the other two β -chains are of another type.
- **Fetal hemoglobin** is composed of **two α subunits** and **two γ (gamma)** subunits. Because of that fetal hemoglobin **has a greater affinity for oxygen** than does maternal hemoglobin and oxygen transfer occurs efficiently across the placenta from the mother to the fetus.
- **Myoglobin**, consist of a **single polypeptide chain**. myoglobin is the **oxygen storage of skeletal muscle** has the prosthetic group called **heme group** which is the site of oxygen binding.
- **Myoglobin** has a **greater affinity** for oxygen than does **hemoglobin**, and so oxygen is efficiently transferred from hemoglobin in the blood to the myoglobin in the tissue.

Denaturation of proteins

- **Denaturation** is a process in which proteins **lose** their **structure**, it occurs when there is a disruption of any of the bonds that stabilize secondary, tertiary or quaternary structure **by** some of external compounds such as **treatment** of proteins with strong acids or bases, **high concentrations of inorganic salts, organic solvents** (e.g., **alcohol, acetone** or **chloroform**), or **heat**.
- The comparatively weak forces responsible for maintaining secondary, tertiary and quaternary structure of proteins are readily disrupted with resulting loss of biologic activity. **This disruption** of native structure is **termed denaturation**.
- Physically, denaturation may be viewed as randomizing the conformation of a polypeptide chain without affecting its primary structure (peptide bonds).
- Most biological proteins lose their biological function when denatured. For example, enzymes lose their activity when denatured.
- Solutions such as **6 M aqueous urea** ($\text{H}_2\text{N}-\text{CO}-\text{NH}_2$), break hydrogen bonds and cause the unfolding of globular proteins.
- Other chemical agents such as **alcohol** also denature proteins, coagulating them. This process is used in sterilizing the skin before injections. At a concentration of **70%, ethanol** penetrates bacteria and kills them by coagulating their proteins, whereas 95% alcohol denatures only surface proteins.
- In living cells, some denaturation caused by heat can be reversed by chaperones. These proteins help a partially heat-denatured protein to regain its native secondary, tertiary, and quaternary structures.
- Some denaturation, however, is **irreversible**. For example we cannot reverse a hard-boiled egg.

Treatment of straight and curly hair:

- **Reducing agents**, such as **2-mercaptoethanol** ($\text{HOCH}_2\text{CH}_2\text{SH}$), can break the (-S-S-) disulfide bonds, reducing them to -SH groups. The processes of permanent waving and straightening of curly hair are examples of the latter effect.
- The protein **keratin**, which makes up human hair, contains a high percentage of disulfide bonds. These bonds are primarily responsible for the shape of the hair, whether straight or curly.
- In either permanent waving or straightening, the hair is first treated with a reducing agent that cleaves some of the (-S-S-) bonds. This treatment allows the molecules to lose their rigid orientations and become more flexible.
- The hair is then set into the desired shape, using curlers or rollers, and an oxidizing agent is applied. The **oxidizing agent** reverses the preceding reaction, forming new disulfide bonds, which now hold the molecules together in the desired positions.

Classification of proteins on the basis of shape and size

1- **Fibrous proteins:** When the axial ratio of length: width of a protein molecule is more than 10, it is called a fibrous protein.

- Examples: **α -keratin** and **collagen**.

2- **Globular protein:** When the axial ratio of length: width of a protein molecule is less than 10, it is called as globular protein.

- Examples: **Myoglobin**, **haemoglobin**, **ribonuclease**, etc.

Classification of proteins on the basis of functional properties

1- Structural proteins: Example\\ Collagen, elastin, Keratin

- Collagen and elastin that provide a fibrous framework in animal connective tissues.

2- Storage protein: Example\\ Casein (the protein of milk), iron is stored as ferritin.

3- Transport proteins: Example\\ Hemoglobin, myoglobin

4- Hormonal proteins: Example\\ Insulin

5- Receptor proteins: Example\\ Membrane receptor proteins

6- Contractile proteins: Example\\ Actin, myosin, cilia, flagella

7- Defenitive proteins: Example\\ Antibodies are known as immunoglobulin

8- Enzymatic proteins: Example\\ Digestive enzymes hydrolyses the polymers in food.

Classification of proteins on the basis of solubility and physical properties

1. Simple proteins:

- These are proteins which on complete hydrolysis yield only amino acids.

2. Conjugated proteins:

- These are proteins which in addition to amino acids contain a non-protein group called prosthetic group in their structure.

3. Derived proteins:

- These are the proteins formed from native protein by the action of heat, physical forces or chemical factors.

Classification of proteins according to its composition:-

I- Simple proteins:

i.e. on hydrolysis gives only amino acids

Examples:

1- Albumin and globulins: present in egg, milk and blood

They are proteins of high biological value i.e. contain all essential amino acids and easily digested.

Types of globulins: **α 1 globulin:** e.g. antitrypsin

α 2 globulin: e.g. hepatoglobulin: protein that binds hemoglobin to prevent its excretion by the kidney

β -globulin: e.g. transferrin: protein that transport iron

E- Elastin: present in walls of large blood vessels (such as aorta). It is very important in lungs, elastic ligaments, skin, cartilage, ..

It is elastic fiber that can be stretched to several times as its normal length.

Structure: composed of 4 polypeptide chains (tetramer), similar to collagen being having 33% glycine and rich in proline but in that it has low hydroxyproline and absence of hydroxylysine.

i.e. On hydrolysis, give protein part and non protein part and subclassified into:

1- Phosphoproteins: These are proteins conjugated with phosphate group. Phosphorus is attached to OH group of serine or threonine.

e.g. Casein of milk and vitellin of yolk.

2- Lipoproteins:

These are proteins conjugated with lipids.

Functions:

- a- help lipids to transport in blood
- b- Enter in cell membrane structure helping lipid soluble substances to pass through cell membranes.

glycoproteins
proteins conjugated with sugar (carbohydrate)

e.g. – Mucin

- Some hormones such as erythropoietin
- present in cell membrane structure
- blood groups.

4- Nucleoproteins: These are basic proteins (e.g. histones) conjugated with nucleic acid (DNA or RNA).

- e.g.
- a- chromosomes: are proteins conjugated with DNA
 - b- Ribosomes: are proteins conjugated with RNA

5- Metalloproteins: These are proteins conjugated with metal like iron, copper, zinc,

a- Iron-containing proteins: Iron may present in heme such as in - hemoglobin (Hb)

- myoglobin (protein of skeletal muscles and cardiacmuscle),
- cytochromes,
- catalase, peroxidases (destroy H_2O_2)
- tryptophan pyrrolase (desrtroy indole ring of tryptophan).

Iron may be present in free state (not in heme) as in:

- Ferritin: Main store of iron in the body. ferritin is present in liver, spleen and bone marrow.
- Hemosidrin: another iron store.
- Transferrin: is the iron carrier protein in plasma.

b- Copper containing proteins:

- Ceruloplasmin which oxidizes ferrous ions into ferric ions.
- Oxidase enzymes such as cytochrome oxidase.

c- Zn containing proteins: e.g. Insulin and carbonic anhydrase.

d- Mg containing proteins:e.g. Kinases and phosphatases.

6-Chromoproteins: These are proteins conjugated with pigment. e.g.

- All proteins containing heme (Hb, myoglobin,)
- Melanoprotein:e.g proteins of hair or iris which contain melanin.

3- Derived proteins:- These are proteins derived by partial to complete hydrolysis from the simple or conjugated proteins by the action of acids, alkalies or enzymes.. example:- fibrin from fibrinogen ,cooked meat and cooked albumin.

Gelatin: from hydrolysis of collagen

((When collagen is heated with water or dil. HCl it will be converted into gelatin which is soluble , digestible and used as diet (as jelly)).

Examples of multisubunit enzymes

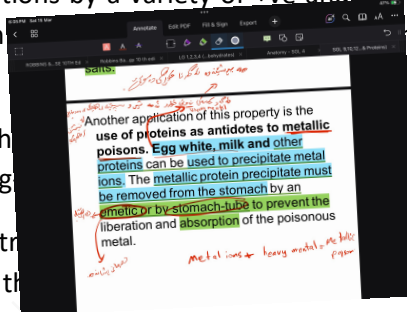
Enzyme	Number of Subunits
HIV protease	2
Hexokinase	2
Bacterial cytochrome oxidase	3
Lactate dehydrogenase	4
Aspartate carbamoyl transferase	12
Human cytochrome oxidase	13

Biologically important proteins

1. **Glutathione** is a **tripeptide** containing **glutamic acid**, **cysteine** and **glycine**. It is present in **erythrocytes** and several other tissues. It acts as a coenzyme and protects haemoglobin against oxidation.
2. **Insulin** and **glucagon** are pancreatic hormones, involved in the regulation of glucose metabolism.
3. **Angiotensin** is a peptide which stimulates the release of certain hormones from adrenal gland and is a vasoconstriction.
- **Angiotensin II** which has (**8 amino acid**) has greater effect on blood pressure.
4. **Albumin** combine with substances of low solubilities such as **cholesterol**, **triacylglycerol** to form more soluble complexes, which can be transported in an aqueous environment of the body fluids. (Human serum albumin contains 584 amino acids.)

Precipitation of proteins:

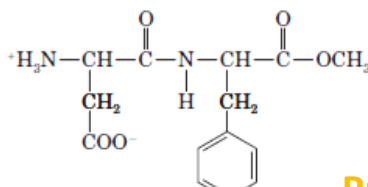
- Proteins can be precipitated from solutions by a variety of +ve and -ve ions. Such precipitation is of importance in the isolation of protein, in urine and other biological fluids and extracts for analysis.
- An example of proteins precipitation: The **Agar** precipitates the proteins of tissues as **Agar**.
- Dentists use a 50% aqueous solution of **trichloroacetic acid**. This strong acid stops the bleeding, kills diseased tissue, and allows the **tooth**.



Another application of this property is the use of proteins as antidotes to metallic poisons. Egg white, milk and other proteins can be used to precipitate metal ions. The metallic protein precipitate must be removed from the stomach by an emetic or by stomach tube to prevent the liberation and absorption of the poisonous metal.

Aspartame (the Sweet Peptide)

- The **dipeptide L-aspartyl-L-phenylalanine** is of considerable commercial importance.
- The soft-drink industry is one of the largest markets for aspartame. **Aspartame is 100 to 150 times sweeter than sucrose**.
- It is made from natural amino acids, so both the **aspartic acid** and the **phenylalanine** have the L configuration. Other possibilities have also been synthesized: the L-D, D-L and D-D configurations. However, they are all bitter rather than sweet.
- You may have noticed that foods containing aspartame carry a warning about the phenylalanine portion of that artificial sweetener.
- This information is of vital importance to people who have phenylketonuria, a genetic disease of phenylalanine metabolism.
- A substitute for aspartame, which carries the trade name **Alatame**, contains alanine rather than phenylalanine. It has been introduced to retain the benefits of aspartame without the dangers associated with phenylalanine.



Biologically important peptides or proteins:

1. Glutathione is a tripeptide containing glutamic acid, cysteine and glycine. It is present in erythrocytes and several other tissues. It acts as a coenzyme and protects haemoglobin against oxidation.

Glutathione is an antioxidant in plants and animals is capable of preventing damage to important cellular components caused by reactive oxygen species such as free radicals, peroxides, and heavy metals.

2. Insulin and glucagon are pancreatic hormones, involved in the regulation of glucose metabolism.
3. Albumin combine with substances of low solubilities such as cholesterol, triacylglycerol to form more soluble complexes, which can be transported in an aqueous environment of the body fluids. (it contains 584 amino acids).

4. The enzyme Renin is released from kidneys and acts on Angiotensinogen, of plasma to release a peptide **Angiotensin I** (10 amino acids) which has only a slight effect on blood pressure.

Angiotensin I is then converted to **Angiotensin II** by splitting off two amino acids which has 8 amino acid, has greater effect on blood pressure. Removal of one residue (aspartic acid) from **Angiotensin II** results in formation of **Angiotensin III** (7 amino acids) which plays role in hypertension.

Angiotensin is a peptide hormone that causes vasoconstriction and an increase in blood pressure. It is part of the renin–angiotensin system, which regulates blood pressure.

Angiotensin also stimulates the release of aldosterone from the adrenal cortex to promote sodium retention by the kidneys.